

COMICCRAFTAI

SYSTEM ARCHITECTURE

1. INTRODUCTION

ComicCraftAI follows a modular web application architecture consisting of a frontend, FastAPI backend, generative AI services, image generation service, layout processing module, and PDF exporter.

2. ARCHITECTURE COMPONENTS

USER

The user provides the story requirements through the web interface.

FRONTEND

The frontend is implemented using HTML, CSS, and Jinja2 templates.

FASTAPI BACKEND

FastAPI receives user requests and coordinates the comic generation workflow.

GEMINI API

Gemini is used for generating:

- Comic outline
- Scene descriptions
- Narration
- Dialogue
- Image prompts

HUGGING FACE INFERENCE API

Hugging Face is used to generate the illustrations for the comic panels.

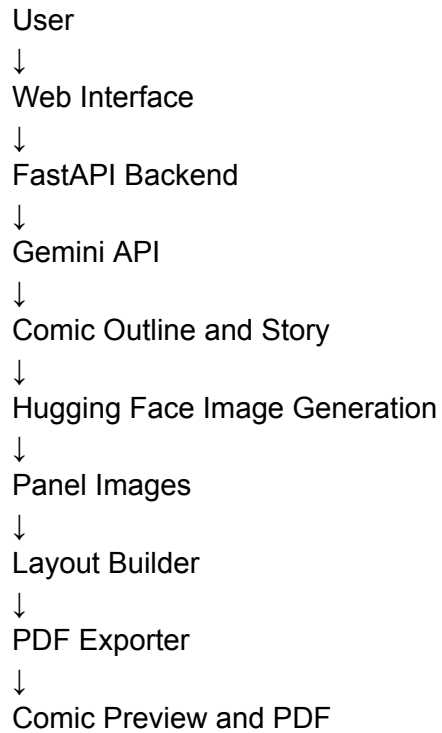
LAYOUT BUILDER

The layout builder combines the generated outline, story information, and images into a structured comic layout.

PDF EXPORTER

The PDF exporter converts the generated comic layout into a downloadable PDF.

3. DATA FLOW



4. ADVANTAGES OF THE ARCHITECTURE

The modular architecture makes the application easier to maintain and allows individual components to be modified or replaced independently.

5. CONCLUSION

The architecture provides a clear separation between user interface, backend processing, AI services, image generation, layout processing, and PDF generation.